



# Cambridge IGCSE™

## CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet  
Soft clean eraser  
Soft pencil (type B or HB is recommended)

## INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

## INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages.



- 1 A gas is placed in a sealed container. The gas has a pressure of one atmosphere and a temperature of 50 °C.

It is heated to 100 °C.

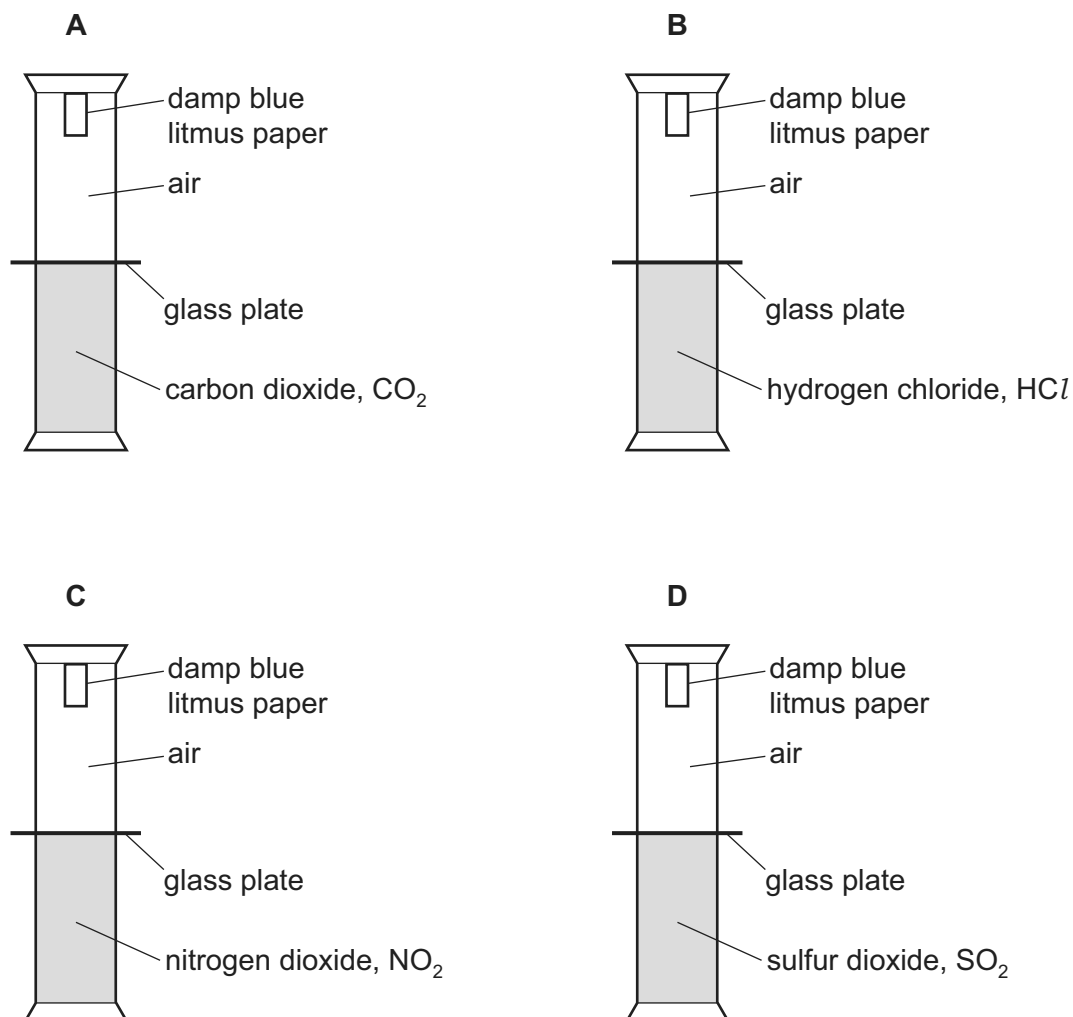
Which row describes the cause of the pressure of the gas and the effect of increasing the temperature of the gas?

|          | cause of gas pressure                              | the effect of increased temperature of the gas   |
|----------|----------------------------------------------------|--------------------------------------------------|
| <b>A</b> | collisions between gas particles                   | collisions become less frequent                  |
| <b>B</b> | collisions between gas particles                   | the average speed of the gas particles increases |
| <b>C</b> | collisions between gas particles and the container | collisions become less frequent                  |
| <b>D</b> | collisions between gas particles and the container | the average speed of the gas particles increases |

- 2 Four experiments, each containing a different acidic gas, are set up as shown.

The dividing glass plates are removed at the same time.

In which set of apparatus does the litmus turn red first?



- 3 The Group I element potassium forms an ionic bond with the Group VII element fluorine.

Which two ions are produced?

- A**  $\text{K}^+$  and  $\text{F}^+$       **B**  $\text{K}^+$  and  $\text{F}^-$       **C**  $\text{K}^-$  and  $\text{F}^-$       **D**  $\text{K}^-$  and  $\text{F}^+$

4 X and Y are atoms.

- X and Y have the same number of electron shells.
- X and Y have the same number of outer electrons.
- X and Y have different mass numbers.

Which statements about X and Y are correct?

- 1 X and Y are isotopes.
- 2 X and Y have the same total number of electrons.
- 3 X and Y have the same chemical properties.

**A** 1, 2 and 3      **B** 1 and 2 only      **C** 1 and 3 only      **D** 2 and 3 only

5 Lithium chloride is an ionic compound and silicon(IV) oxide is a covalent compound.

Which statement about **both** compounds is correct?

- A** They are not soluble in water.
- B** They conduct electricity when melted.
- C** They do not conduct electricity in solid form.
- D** They have low melting points.

6 Which equations are balanced?

- 1  $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
- 2  $\text{ZnCO}_3 + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{CO}_2 + 2\text{H}_2\text{O}$
- 3  $\text{Mg}(\text{NO}_3)_2 + \text{NaOH} \rightarrow \text{Mg}(\text{OH})_2 + 2\text{NaNO}_3$
- 4  $\text{CaCO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + \text{H}_2\text{O} + \text{CO}_2$

**A** 1 and 2      **B** 1 and 4      **C** 2 and 3      **D** 3 and 4

7 Which row shows the formulae of sodium carbonate, zinc nitrate and ammonium sulfate?

|          | sodium carbonate         | zinc nitrate               | ammonium sulfate             |
|----------|--------------------------|----------------------------|------------------------------|
| <b>A</b> | $\text{Na}_2\text{CO}_3$ | $\text{ZnNO}_3$            | $(\text{NH}_4)_2\text{SO}_4$ |
| <b>B</b> | $\text{Na}_2\text{CO}_3$ | $\text{Zn}(\text{NO}_3)_2$ | $(\text{NH}_4)_2\text{SO}_4$ |
| <b>C</b> | $\text{NaCO}_3$          | $\text{ZnNO}_3$            | $(\text{NH}_3)_2\text{SO}_4$ |
| <b>D</b> | $\text{NaCO}_3$          | $\text{Zn}(\text{NO}_3)_2$ | $(\text{NH}_3)_2\text{SO}_4$ |

8 Which statements about hydrogen and oxygen are correct?

|          | hydrogen and oxygen<br>can react to produce<br>electrical energy | hydrogen and oxygen can<br>be made by the electrolysis of<br>dilute aqueous sodium chloride |
|----------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>A</b> | x                                                                | x                                                                                           |
| <b>B</b> | x                                                                | ✓                                                                                           |
| <b>C</b> | ✓                                                                | x                                                                                           |
| <b>D</b> | ✓                                                                | ✓                                                                                           |

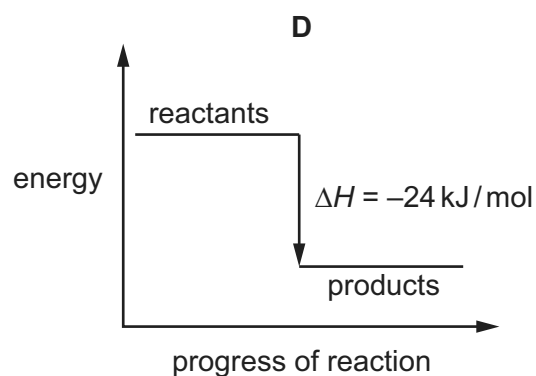
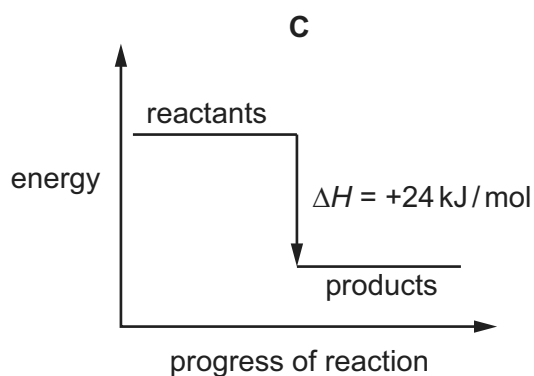
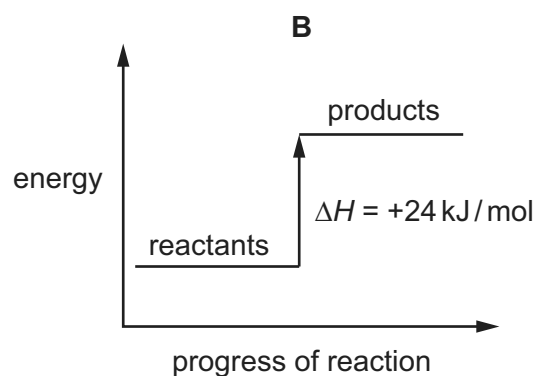
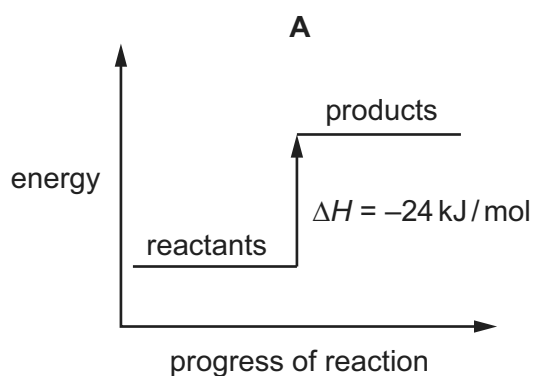
9 Graphite has a giant covalent structure.

Which statements about graphite are correct?

- 1 Carbon atoms form four covalent bonds with neighbouring atoms.
- 2 There are delocalised electrons between layers of carbon atoms.
- 3 Graphite is a useful lubricant.
- 4 Graphite is a good conductor of electricity.

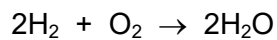
**A** 1 and 2      **B** 1, 3 and 4      **C** 2, 3 and 4      **D** 3 and 4 only

10 Which reaction pathway diagram represents an endothermic reaction?



**11** Hydrogen burns in oxygen.

The equation for the reaction is shown.



The table shows the bond energies involved.

| bond | bond energy<br>in kJ/mol |
|------|--------------------------|
| H–H  | 436                      |
| O=O  | 498                      |
| O–H  | 464                      |

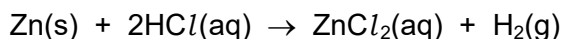
What is the energy given out during the reaction?

- A** –3226 kJ/mol
- B** –884 kJ/mol
- C** –486 kJ/mol
- D** –442 kJ/mol

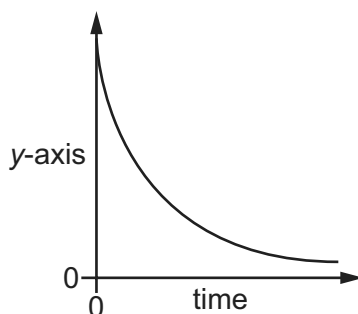
**12** Which process involves a chemical change?

- A** adding sodium to water
- B** boiling water
- C** dissolving sodium chloride in water
- D** producing water from aqueous sodium chloride

- 13 An experiment is carried out to find the rate of reaction between hydrochloric acid and zinc.

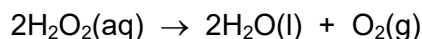


The results of the experiment are shown.



What is the label on the y-axis?

- A amount of  $\text{ZnCl}_2$  produced
  - B concentration of  $\text{HCl}$
  - C mass of  $\text{Zn}$  reacted
  - D volume of  $\text{H}_2$  produced
- 14 Hydrogen peroxide,  $\text{H}_2\text{O}_2$ , decomposes to form water and oxygen.



Manganese(IV) oxide catalyses the decomposition reaction.

The reaction is investigated in four experiments.

| experiment | volume and concentration of hydrogen peroxide   | conditions                                  |
|------------|-------------------------------------------------|---------------------------------------------|
| 1          | 12.5 cm <sup>3</sup> of 1.0 mol/dm <sup>3</sup> | 25 °C with manganese(IV) oxide powder added |
| 2          | 12.5 cm <sup>3</sup> of 2.0 mol/dm <sup>3</sup> | 40 °C with manganese(IV) oxide powder added |
| 3          | 25 cm <sup>3</sup> of 1.0 mol/dm <sup>3</sup>   | 40 °C without manganese(IV) oxide powder    |
| 4          | 25 cm <sup>3</sup> of 1.0 mol/dm <sup>3</sup>   | 40 °C with manganese(IV) oxide powder added |

All reactions go to completion and all measurements of gas volumes are at room temperature and pressure.

Which statement is correct?

- A Experiment 1 produces less gas than experiment 4, but at the same rate.
- B Experiment 2 produces more gas than experiment 1, but at the same rate.
- C Experiment 2 and experiment 4 each produce the same volume of gas, but at different rates.
- D Experiment 3 and experiment 4 each produce the same volume of gas and at the same rate.

**15** Sulfuric acid is produced by the Contact process.

Which row shows the typical conditions used in the process?

|          | catalyst          | pressure<br>/ kPa | temperature<br>/ °C |
|----------|-------------------|-------------------|---------------------|
| <b>A</b> | iron              | 200               | 300                 |
| <b>B</b> | iron              | 20 000            | 450                 |
| <b>C</b> | vanadium(V) oxide | 200               | 450                 |
| <b>D</b> | vanadium(V) oxide | 20 000            | 300                 |

**16** Which equation shows the reduction of copper?

- A**  $\text{CuO} + \text{C} \rightarrow \text{Cu} + \text{CO}$
- B**  $2\text{CuS} + 3\text{O}_2 \rightarrow 2\text{CuO} + 2\text{SO}_2$
- C**  $\text{Cu(g)} \rightarrow \text{Cu(l)}$
- D**  $\text{Cu(l)} \rightarrow \text{Cu(s)}$

**17** Which statement about acids is correct?

- A** A weak acid partially dissociates in aqueous solution.
- B** An acid accepts protons when added to water.
- C** Ethanoic acid acts as a strong acid when added to water.
- D** Hydrochloric acid is a strong acid that ionises in water to form  $\text{H}^+$  ions.

**18** Copper(II) sulfate is formed by reacting excess solid copper(II) carbonate with dilute sulfuric acid.

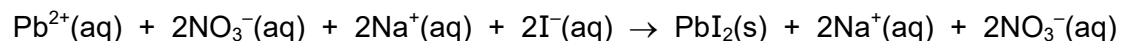
Which processes are part of the preparation of solid copper(II) sulfate?

- 1 crystallisation
- 2 distillation
- 3 filtration
- 4 titration

- A** 1 and 3      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4



19 Which type of reaction is represented by the equation shown?



- A addition
- B redox
- C neutralisation
- D precipitation

20 Which compound is likely to be coloured?

- A  $\text{KMnO}_4$       B  $\text{KNO}_3$       C  $\text{K}_2\text{CO}_3$       D  $\text{K}_2\text{SO}_4$

21 Which statements about the metal zinc are correct?

- 1 It is extracted from the ore bauxite.
- 2 It is used to galvanise steel.
- 3 It is used to make the alloy brass.
- 4 It reacts with dilute hydrochloric acid to produce hydrogen gas.

- A 1, 2 and 4      B 1, 3 and 4      C 2, 3 and 4      D 2 and 3 only

22 The electronic configurations of four elements, P, Q, R and S, are shown.

| element | electronic configuration |
|---------|--------------------------|
| P       | 2                        |
| Q       | 2,2                      |
| R       | 2,6                      |
| S       | 2,8                      |

Which elements are unreactive monatomic gases?

- A P and Q      B P and S      C Q and R      D S only

- 23** Which row compares the strength of alloys with pure metals and explains the difference in strength?

|          | strength of an alloy compared to a pure metal | explanation                                                      |
|----------|-----------------------------------------------|------------------------------------------------------------------|
| <b>A</b> | weaker                                        | larger atoms slide more easily over smaller atoms                |
| <b>B</b> | weaker                                        | larger atoms make it harder for layers to slide over one another |
| <b>C</b> | stronger                                      | larger atoms slide more easily over smaller atoms                |
| <b>D</b> | stronger                                      | larger atoms make it harder for layers to slide over one another |

- 24** Zinc oxide reacts with carbon to produce zinc.

Which equation represents this reaction?

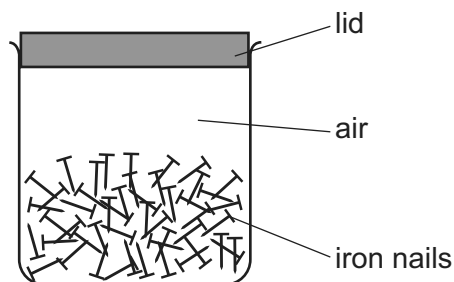
- A**  $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}$
- B**  $2\text{ZnO} + 2\text{C} \rightarrow 2\text{Zn} + 2\text{CO}_2$
- C**  $\text{ZnO} + \text{C} \rightarrow \text{Zn} + \text{CO}$
- D**  $\text{ZnO} + 2\text{C} \rightarrow \text{Zn} + 2\text{CO}_2$

- 25** When a piece of aluminium foil is added to dilute hydrochloric acid, no effervescence is seen.

Which statement explains why no effervescence is seen?

- A** Aluminium does not make a gas when it reacts with an acid.
- B** Aluminium has a surface layer of aluminium oxide.
- C** Aluminium is less reactive than hydrogen.
- D** Aluminium only reacts with concentrated acid.

26 Iron nails are stored in an airtight container.



The nails begin to rust after a few days.

How can the rusting of the nails be prevented?

- A Leave the lid off.
- B Replace the air with argon.
- C Put the container in a warm place.
- D Seal the container in a bag.

27 Four substances present in the blast furnace during iron extraction are listed.

- 1 calcium carbonate
- 2 carbon dioxide
- 3 carbon monoxide
- 4 iron(III) oxide

Which substances are both a reactant and a product during the reactions occurring in the blast furnace?

- A 1 and 2
- B 1 and 4
- C 2 and 3
- D 3 and 4

28 Aluminium is extracted from purified bauxite by electrolysis.

Which row shows the ionic half-equations for the reaction at each electrode?

|   | anode                              | cathode                            |
|---|------------------------------------|------------------------------------|
| A | $Al \rightarrow Al^{3+} + 3e^{-}$  | $2O^{2-} + 4e^{-} \rightarrow O_2$ |
| B | $Al^{3+} + 3e^{-} \rightarrow Al$  | $2O^{2-} \rightarrow O_2 + 4e^{-}$ |
| C | $2O^{2-} + 4e^{-} \rightarrow O_2$ | $Al \rightarrow Al^{3+} + 3e^{-}$  |
| D | $2O^{2-} \rightarrow O_2 + 4e^{-}$ | $Al^{3+} + 3e^{-} \rightarrow Al$  |

29 Which test is used to show that a sample of water is pure?

- A Evaporate the water to see if any solids remain.
- B Heat the water to check its boiling point.
- C Test with anhydrous cobalt(II) chloride.
- D Use universal indicator paper to check its pH.

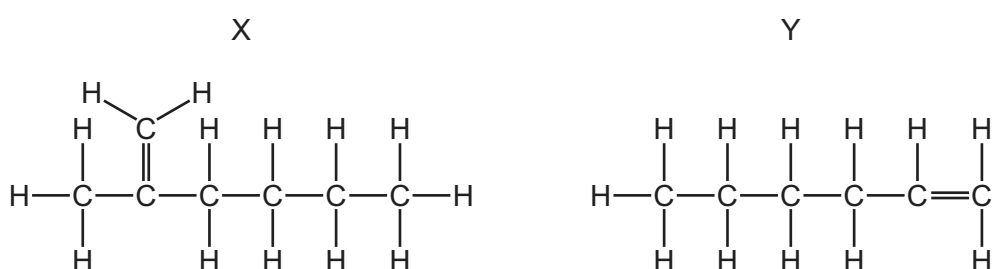
30 Catalytic converters in car exhausts change polluting gases into non-polluting gases.

Which statements about oxides of nitrogen and car engines are correct?

- 1 The nitrogen in oxides of nitrogen comes from compounds in gasoline.
- 2 The oxygen in oxides of nitrogen comes from the air in the car engine.
- 3 Catalytic converters convert oxides of nitrogen into nitrogen.

- A 1 and 2      B 2 and 3      C 2 only      D 3 only

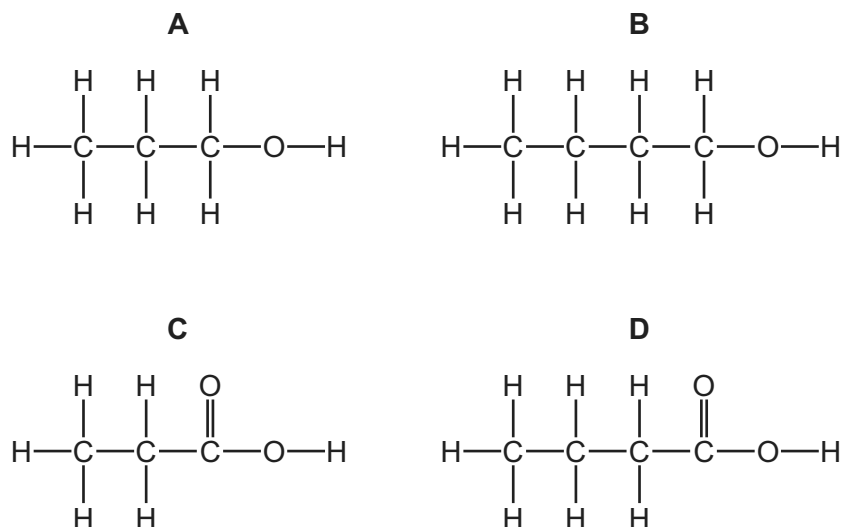
31 The structures of two molecules, X and Y, are shown.



Which row describes X and Y?

|   | structural isomers | belong to same homologous series |
|---|--------------------|----------------------------------|
| A | no                 | no                               |
| B | no                 | yes                              |
| C | yes                | no                               |
| D | yes                | yes                              |

32 What is the structure of butanoic acid?



33 When a mixture of methane and chlorine is exposed to ultraviolet light, a reaction takes place.

Which statements about this reaction are correct?

- 1 It is an addition reaction.
- 2 The ultraviolet light provides the activation energy.
- 3 An equation for the reaction is  $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{H}_2$ .
- 4  $\text{CH}_3\text{Cl}$  is made in the reaction.

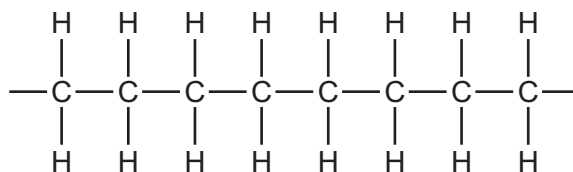
**A** 1 and 3      **B** 1 and 4      **C** 2 and 3      **D** 2 and 4

34 Esters are formed when a carboxylic acid reacts with an alcohol.

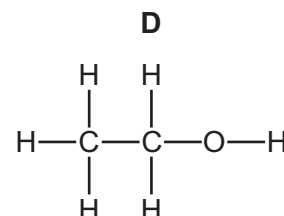
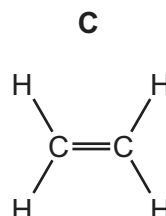
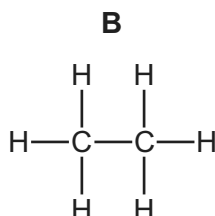
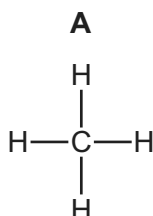
What is the catalyst for this reaction?

- A** aqueous potassium manganate(VII)  
**B** iron  
**C** sulfuric acid  
**D** vanadium(V) oxide

35 The diagram shows part of a polymer.



Which diagram shows the monomer from which this polymer is made?



36 Nylon and PET are polymers.

Which statements about these polymers are correct?

- 1 They are both condensation polymers.
- 2  $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$  could be a monomer for both polymers.
- 3 The complete combustion of both polymers gives two products only.

**A** 1 and 2      **B** 1 and 3      **C** 1 only      **D** 2 and 3

37 Ethane is used as a fuel.

Which equation shows the complete combustion of ethane?

- A**  $2\text{C}_2\text{H}_6 + 7\text{O}_2 \rightarrow 4\text{CO}_2 + 6\text{H}_2\text{O}$
- B**  $2\text{C}_2\text{H}_6 + 5\text{O}_2 \rightarrow 4\text{CO} + 6\text{H}_2\text{O}$
- C**  $\text{C}_2\text{H}_4 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$
- D**  $\text{C}_2\text{H}_4 + 2\text{O}_2 \rightarrow 2\text{CO} + 2\text{H}_2\text{O}$

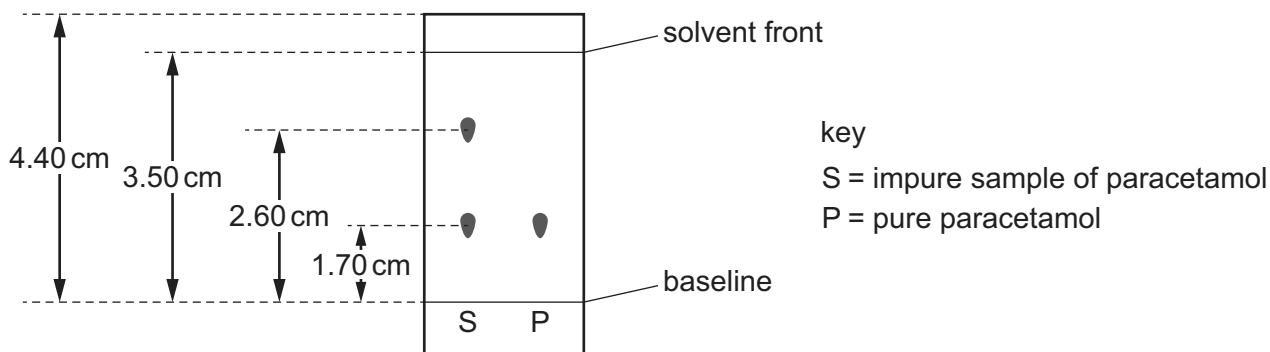
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- 38 The painkiller paracetamol is synthesised from 4-aminophenol.

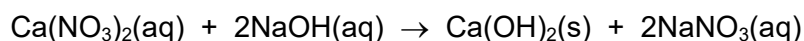
Chromatography is done on an impure sample of paracetamol. The results are shown. The diagram is not drawn to scale.



The sample of paracetamol is contaminated with 4-aminophenol only.

What is the  $R_f$  value of 4-aminophenol?

- A** 0.49                      **B** 0.65                      **C** 0.74                      **D** 1.35
- 39 The equation for the reaction of aqueous calcium nitrate and aqueous sodium hydroxide is shown.



Which process is used to remove calcium hydroxide from the mixture?

- A** chromatography  
**B** crystallisation  
**C** distillation  
**D** filtration
- 40 The results of two tests on aqueous compound X are given.

| test                                                  | result              |
|-------------------------------------------------------|---------------------|
| warm with aluminium foil and aqueous sodium hydroxide | ammonia is produced |
| aqueous sodium hydroxide                              | brown precipitate   |

What is X?

- A** iron(III) nitrate  
**B** iron(II) nitrate  
**C** iron(III) sulfate  
**D** iron(II) sulfate

## The Periodic Table of Elements

| Group                      |                             |                                                                |                             |                             |                              |                              |                              |                            |                              |                            |                            |                           |                            |                             |                              |                          |                          |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
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| I                          | II                          | Key                                                            |                             |                             |                              |                              |                              |                            |                              |                            |                            | III                       | IV                         | V                           | VI                           | VII                      | VIII                     |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
| 3<br>Li<br>lithium<br>7    | 4<br>Be<br>beryllium<br>9   | atomic number<br>atomic symbol<br>name<br>relative atomic mass |                             |                             |                              |                              |                              |                            |                              |                            |                            | 5<br>B<br>boron<br>11     | 6<br>C<br>carbon<br>12     | 7<br>N<br>nitrogen<br>14    | 8<br>O<br>oxygen<br>16       | 9<br>F<br>fluorine<br>19 | 2<br>He<br>helium<br>4   |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
|                            |                             | 1<br>H<br>hydrogen<br>1                                        |                             |                             |                              |                              |                              |                            |                              |                            |                            |                           |                            |                             |                              |                          |                          |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
|                            |                             |                                                                |                             |                             |                              |                              |                              |                            |                              |                            |                            |                           |                            |                             |                              |                          |                          |                            |                           |                           |                             |                             |                            |                            |                           |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
| 11<br>Na<br>sodium<br>23   | 12<br>Mg<br>magnesium<br>24 | 13<br>Al<br>aluminium<br>27                                    | 14<br>Si<br>silicon<br>28   | 15<br>P<br>phosphorus<br>31 | 16<br>S<br>sulfur<br>32      | 17<br>Cl<br>chlorine<br>35.5 | 18<br>Ar<br>argon<br>40      | 19<br>K<br>potassium<br>39 | 20<br>Ca<br>calcium<br>40    | 21<br>Sc<br>scandium<br>45 | 22<br>Ti<br>titanium<br>48 | 23<br>V<br>vanadium<br>51 | 24<br>Cr<br>chromium<br>52 | 25<br>Mn<br>manganese<br>55 | 26<br>Fe<br>iron<br>56       | 27<br>Co<br>cobalt<br>59 | 28<br>Ni<br>nickel<br>59 | 29<br>Cu<br>copper<br>64   | 30<br>Zn<br>zinc<br>65    | 31<br>Ga<br>gallium<br>70 | 32<br>Ge<br>germanium<br>73 | 33<br>As<br>arsenic<br>75   | 34<br>Se<br>selenium<br>79 | 35<br>Br<br>bromine<br>80  | 36<br>Kr<br>krypton<br>84 |                            |                             |                         |                            |                             |                         |                            |                           |                           |                        |                           |                         |                     |                                 |                           |                              |                           |                           |                              |                                |                               |                               |                            |                             |                             |                               |                              |                             |
| 37<br>Rb<br>rubidium<br>85 | 38<br>Sr<br>strontium<br>88 | 39<br>Y<br>yttrium<br>89                                       | 40<br>Zr<br>zirconium<br>91 | 41<br>Nb<br>niobium<br>93   | 42<br>Mo<br>molybdenum<br>96 | 43<br>Tc<br>technetium<br>—  | 44<br>Ru<br>ruthenium<br>101 | 45<br>Rh<br>rhodium<br>103 | 46<br>Pd<br>palladium<br>106 | 47<br>Ag<br>silver<br>108  | 48<br>Cd<br>cadmium<br>112 | 49<br>In<br>indium<br>115 | 50<br>Sn<br>tin<br>119     | 51<br>Sb<br>antimony<br>122 | 52<br>Te<br>tellurium<br>128 | 53<br>I<br>iodine<br>127 | 54<br>Xe<br>xenon<br>131 | 55<br>Cs<br>caesium<br>133 | 56<br>Ba<br>barium<br>137 | 57–71<br>lanthanoids      | 72<br>Hf<br>hafnium<br>178  | 73<br>Ta<br>tantalum<br>181 | 74<br>W<br>tungsten<br>184 | 75<br>Re<br>rhenium<br>186 | 76<br>Os<br>osmium<br>190 | 77<br>Ir<br>iridium<br>192 | 78<br>Pt<br>platinum<br>195 | 79<br>Au<br>gold<br>197 | 80<br>Hg<br>mercury<br>201 | 81<br>Tl<br>thallium<br>204 | 82<br>Pb<br>lead<br>207 | 83<br>Bi<br>bismuth<br>209 | 84<br>Po<br>polonium<br>— | 85<br>At<br>astatine<br>— | 86<br>Rn<br>radon<br>— | 87<br>Fr<br>francium<br>— | 88<br>Ra<br>radium<br>— | 89–103<br>actinoids | 104<br>Rf<br>rutherfordium<br>— | 105<br>Db<br>dubnium<br>— | 106<br>Sg<br>seaborgium<br>— | 107<br>Bh<br>bohrium<br>— | 108<br>Hs<br>hassium<br>— | 109<br>Mt<br>meitnerium<br>— | 110<br>Ds<br>darmstadtium<br>— | 111<br>Rg<br>roentgenium<br>— | 112<br>Cn<br>copernicium<br>— | 113<br>Nh<br>nihonium<br>— | 114<br>Fl<br>flerovium<br>— | 115<br>Mc<br>moscovium<br>— | 116<br>Lv<br>livermorium<br>— | 117<br>Ts<br>tennessine<br>— | 118<br>Og<br>oganesion<br>— |

|             |           |    |           |          |    |    |        |         |     |    |              |              |     |    |           |         |     |    |            |           |    |    |          |           |    |    |          |           |    |    |            |        |    |    |         |           |    |    |            |             |    |    |         |             |    |     |        |         |    |     |         |             |    |     |           |          |    |     |          |            |   |
|-------------|-----------|----|-----------|----------|----|----|--------|---------|-----|----|--------------|--------------|-----|----|-----------|---------|-----|----|------------|-----------|----|----|----------|-----------|----|----|----------|-----------|----|----|------------|--------|----|----|---------|-----------|----|----|------------|-------------|----|----|---------|-------------|----|-----|--------|---------|----|-----|---------|-------------|----|-----|-----------|----------|----|-----|----------|------------|---|
| lanthanoids | 57        | La | lanthanum | 139      | 58 | Ce | cerium | 140     | 59  | Pr | praseodymium | 141          | 60  | Nd | neodymium | 144     | 61  | Pm | promethium | —         | 62 | Sm | samarium | 150       | 63 | Eu | europium | 152       | 64 | Gd | gadolinium | 157    | 65 | Tb | terbium | 159       | 66 | Dy | dysprosium | 163         | 67 | Ho | holmium | 165         | 68 | Er  | erbium | 167     | 69 | Tm  | thulium | 169         | 70 | Yb  | ytterbium | 173      | 71 | Lu  | lutetium | 175        |   |
|             | actinoids | 89 | Ac        | actinium | —  | 90 | Th     | thorium | 232 | 91 | Pa           | protactinium | 231 | 92 | U         | uranium | 238 | 93 | Np         | neptunium | —  | 94 | Pu       | plutonium | —  | 95 | Am       | americium | —  | 96 | Cm         | curium | —  | 97 | Bk      | berkelium | —  | 98 | Cf         | californium | —  | 99 | Es      | einsteinium | —  | 100 | Fm     | fermium | —  | 101 | Md      | mendelevium | —  | 102 | No        | nobelium | —  | 103 | Lr       | lawrencium | — |

The volume of one mole of any gas is  $24\text{ dm}^3$  at room temperature and pressure (r.t.p.).